

11.3.5★ Let x and y be numbers such that

$$|x| \neq |y|, \quad x^3 = 15x + 4y, \quad \text{and} \quad y^3 = 4x + 15y.$$

In this problem we find $x^2 + y^2$. (Source: *Dropped AIME*)

11.4.4 Rationalize the denominator of $\frac{2}{\sqrt{2} - \sqrt{5} + \sqrt{7}}$.

Problem 11.25: How many ordered pairs (m, n) of positive integers are solutions to

$$\frac{4}{m} + \frac{2}{n} = 1?$$

(Source: AHSME)

11.43 Three consecutive positive odd integers a , b , and c satisfy $b^2 - a^2 = 344$ and $c^2 - b^2 > 0$. What is the value of $c^2 - b^2$?

11.45 Compute the product

$$\frac{(1998^2 - 1996^2)(1998^2 - 1995^2) \cdots (1998^2 - 0^2)}{(1997^2 - 1996^2)(1997^2 - 1995^2) \cdots (1997^2 - 0^2)}.$$

11.52 A triangle is a right triangle if the sum of the squares of its two smallest sides equals the square of its largest side. If n is a positive integer, must the triangle with sides

$$6(10^{n+2}), 1125(10^{2n+1}) - 8, \text{ and } 1125(10^{2n+1}) + 8$$

be a right triangle? (Source: AHSME) Hints: 75

11.53★ Determine $\sqrt{1 + 50 \cdot 51 \cdot 52 \cdot 53}$ without a calculator. (Source: *MATHCOUNTS*) **Hints:** 109, 127

11.54★ Factor $x^4 + 4y^4$. **Hints:** 83, 135

11.56★ Simplify the expression

$$\frac{(\sqrt{2} - 1)^{1-\sqrt{3}}}{(\sqrt{2} + 1)^{1+\sqrt{3}}},$$

writing your answer as $a - b\sqrt{c}$, where a , b , and c are positive integers. (Source: Mandelbrot)

11.57★ Without a calculator, find the sum of the digits of the number $2003^4 - 1997^4$. (Source: Mandelbrot)

Problem 13.16: In this problem we find all the values of m for which the roots of $2x^2 - mx - 8 = 0$ differ by $m - 1$. (Source: HMMT)

- (a) Find an expression in terms of a , b , and c for the difference between the roots of the quadratic equation $ax^2 + bx + c = 0$.
- (b) Solve the given problem.

13.30 Show that if $c \neq 0$, then the sum of the reciprocals of the roots of the quadratic equation $ax^2 + bx + c = 0$ is $-b/c$.

13.32 Find all values of z such that $z^4 - 4z^2 + 3 = 0$

13.37 What ordered pairs satisfy the system

$$\begin{aligned}x + 3y &= 4, \\x^2 + y^2 - 4y &= 12?\end{aligned}$$

13.38 Find all solutions (including solutions that are not real) to the following equations:

(c) $x^3 - 27 = 0$

13.40 The ancient Greeks believed that the most visually pleasing rectangles were those whose length, ℓ , and width, w , satisfied the equation

$$\frac{\ell + w}{\ell} = \frac{\ell}{w}.$$

Find ℓ/w . (This ratio is called the **Golden ratio**, and it appears in mathematics in many surprising places.) Hints: 212

13.41 The solutions of the equation $x^2 + 2x + 2 = 0$ are $x = p$ and $x = q$. Find a quadratic equation with solutions $y = \frac{1}{p}$ and $y = \frac{1}{q}$. **Hints:** 104, 175

13.42* Find all x that satisfy the equation $\frac{x^6 - 8}{x^2 - 2} = 12$. (Source: UNCC) **Hints:** 157

13.43★ Show that the two values

$$x = \frac{2c}{-b \pm \sqrt{b^2 - 4ac}}$$

are the roots of the equation $ax^2 + bx + c = 0$, where a , b , and c are constants, and a and c are nonzero.
Hints: 15, 72, 117

13.45★ Find all values of r such that $(r^2 + 3r)(r^2 + 3r + 5) = 6$. **Hints:** 188

4.1.6 Find the equation of the parabola having its axis of symmetry parallel to the x -axis and passing through the points $(1, 0)$, $(2, 1)$, and $(2, -2)$.

4.2.5* For what values of k does the graph of the equation $x + ky = 4$ meet the graph of the equation $x^2 + y^2 - 12x + 8y + 42 = 0$ at exactly one point? **Hints:** 223

14.16 Graph the equation $x = 2y^2 - 4y + 4$. What is its vertex?

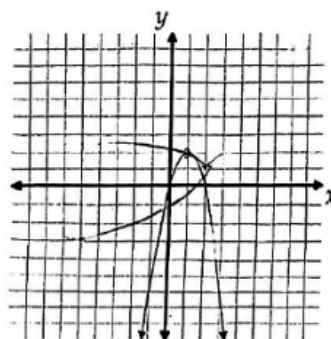
14.20 Find the radius and center of the circle that is the graph of the equation $4x^2 + 4y^2 + 4x - 16y = 7$.

14.23

14.23 Find an equation whose graph is a parabola with axis of symmetry parallel to the x -axis such that the parabola passes through the points $(4, 0)$, $(13, -1)$, and $(4, 2)$.

14.31★ A parabola has its vertex at $(4, -5)$ and meets the x -axis at two points that are on opposite sides of the y -axis. Suppose this parabola is the graph of $y = ax^2 + bx + c$. Determine which of a , b , and c must be positive, which must be negative, and which could be either. (Source: AHSME) Hints: 111, 180

14.33★ If the parabola defined by $y = ax^2 + 6$ is **tangent** to the line $y = x$, then calculate the constant a . (A line is **tangent** to a parabola if it touches the parabola at one point, but is otherwise always “outside” the parabola.) (Source: Mandelbrot)



15.1.3 For what values of r is $2r^2 - 3r > -7$?

Problem 15.7: In this problem, we solve the inequality $\frac{2}{x-2} - \frac{2}{x+3} > -1$.

- (a) Move all terms to one side and write that side with a common denominator.
- (b) For what values of x is the numerator positive? For what values of x is the denominator positive?
- (c) Solve the inequality.

15.2.2 What values of r satisfy the inequality $(r - 3)^2(4r^2 - 25) > 0$?

15.2.3 Solve the inequality $\frac{1}{x-7} - \frac{2x}{1-x} \geq 2$.

15.4.4★ A farmer has 20 meters of fencing to build a chicken run for his chickens. He can use the side of his barn as one side of the chicken run. What is the largest area his chicken run can be if it must be rectangular?

15.22 Solve the inequality $(2x + 4)(-3x^2 - 5x + 2) < 0$.

15.34 Find all ordered triples of integers (x, y, z) that satisfy $(x + 5)^2 + (y - 2)^2 + (z + 3)^2 = 1$. **Hints:** 173

15.44★ Let a be a positive real constant. The minimum value of the expression $2x^2 + ax + 2y^2 - ay + a^2$ is 72. Compute the value of a . **Hints:** 229